

MDI 3003 – Advanced Predictive Analysis

Sarah Blessy N
23MID 0296

LAB Assessment 1

House Price Prediction Using Linear Regression

Industry Style Report

S.No	Title	Page
1	Executive Summary	1
2	Business Problem	1
3	Dataset Description	2
4	Methodology	3
5	EDA and Feature Engineering	3
6	Model Development	4
7	Evaluation Results	4
8	Interpretations	6
9	Limitations And Risks	7
10	Future Improvements	7
11	Conclusion	8

1. Executive Summary

The given experiment demonstrates the process of building and evaluating machine learning regression models for house price prediction. The user imports the required libraries, loads the dataset, performs data inspection, and conducts exploratory data analysis to understand the data. The dataset is then prepared and divided into training and testing sets for model development. Multiple regression algorithms are implemented and evaluated using standard performance metrics. Finally, the user compares the models, improves performance through validation and tuning, and saves the trained model for future use.

2. Business Problem

In the given experiment, the user develops a predictive model to estimate median house prices using housing related features. The primary stakeholders include home buyers, real estate agents, property developers, and financial institutions who rely on accurate price estimates for informed decision making. The target variable is the median house value expressed in US dollars.

The prediction unit is one housing record representing a specific geographical location. Prediction errors may lead to incorrect property valuation, poor investment decisions, or inaccurate loan approvals. The model is intended for educational and analytical purposes and should not be used as the sole basis for legal, financial, or commercial decisions.

3. Dataset Description

The dataset used in this experiment is the California Housing Prices dataset, which is widely used for regression and predictive analytics tasks. The original dataset was developed by R. Kelley Pace and Ronald Barry in their research titled Sparse Spatial Autoregressions (Statistics and Probability Letters, 1997) and was obtained from the StatLib Repository. The version used in this experiment was later modified by Aurélien Géron for the book Hands-On Machine Learning with Scikit-Learn and TensorFlow, where the categorical feature `ocean_proximity` was added.

This version is publicly available through the Kaggle platform.

The dataset represents housing information collected from the 1990 California Census. Each observation corresponds to a single census block group, which is the smallest geographical unit for which the United States Census Bureau publishes sample statistics. The dataset contains 20,640 records with 10 attributes, including both numerical and categorical variables. The target variable is `median_house_value`, expressed in United States Dollars (USD)

Before model development, the user examines the dataset for missing values, duplicate records, data types, and statistical summaries to ensure data quality. The dataset is generally clean and suitable for regression analysis. However, it has certain limitations. The data represents housing conditions only from the year 1990 and may not reflect current market trends. Additionally, the dataset is limited to California and therefore may not generalize to housing markets in other regions.

Attribute	Description
Dataset Name	California Housing Prices
Original Source	Pace, R. Kelley and Ronald Barry (1997), StatLib Repository
Modified Version	Aurélien Géron, <i>Hands-On Machine Learning with Scikit-Learn and TensorFlow</i>
Distribution Source	Kaggle
Coverage	California, United States (1990 Census)
Observation Unit	One census block group
Number of Records	20,640
Number of Attributes	10 (9 input features, 1 target variable)
Target Variable	<code>median_house_value</code> (USD)
Dataset Quality	Clean, structured, minimal preprocessing required
Limitations	Based on 1990 census data and limited to California housing market

4. Methodology

- The user begins by importing the required Python libraries and loading the California Housing dataset into the Jupyter Notebook environment.
- The dataset is inspected for missing values, duplicate records, data types, and descriptive statistics before model development. This preprocessing step ensures that the dataset is suitable for machine learning.
- The user performs Exploratory Data Analysis (EDA) using statistical summaries and visualizations to understand feature distributions and relationships with the target variable.
- The input features are separated from the target variable (`median_house_value`), and the dataset is divided into training and testing sets using an appropriate train test split. This ensures that model evaluation is performed on unseen data.
- Feature preprocessing, such as scaling and categorical variable encoding where required, is applied only to the training data and then reused for the testing data to maintain consistency.
- Multiple regression models, including Dummy Regressor, Linear Regression, Decision Tree Regression, Polynomial Regression, and Random Forest Regression, are trained and compared.
- Model performance is validated using regression metrics such as Mean Absolute Error (MAE), Mean Squared Error (MSE), Root Mean Squared Error (RMSE), and R^2 Score to measure prediction accuracy.
- Cross validation is performed to evaluate model stability across multiple data partitions and to obtain reliable performance estimates.
- Hyperparameter tuning is carried out for suitable models to identify the parameter values that produce the best predictive performance.
- Data leakage is prevented by ensuring that the testing dataset is never used during preprocessing, model training, feature engineering, or hyperparameter tuning. Random seeds are specified wherever applicable to ensure reproducibility, allowing the experiment to produce consistent results across multiple executions.

5. Exploratory Data Analysis [EDA] and Feature Engineering

The user performs Exploratory Data Analysis to understand the characteristics of the California Housing dataset before developing predictive models.

- Histograms reveal that most numerical features exhibit skewed distributions, indicating the presence of non normal data.
- Box plots identify outliers in variables such as `median_income`, `housing_median_age`, and `population`.
- Since tree based models are generally robust to outliers, no outlier removal is performed.
- Correlation analysis shows that `median_income` has the strongest positive relationship with the target variable, making it one of the most influential predictors.
- Scatter plots further confirm that income and geographical location significantly affect house prices.

- The categorical feature `ocean_proximity` is converted into numerical form through encoding so that regression models can process it effectively.
- Feature scaling is applied where appropriate for regression algorithms that are sensitive to feature magnitude, while tree based models are trained without scaling.

These findings guide feature selection, preprocessing decisions, and the choice of regression models for improved prediction accuracy.

6. Model Development

- The user first develops a Dummy Regressor as the baseline model to establish a reference level of prediction performance.
- The baseline model is compared with multiple regression algorithms, including Linear Regression, Decision Tree Regression, Polynomial Regression, and Random Forest Regression.
- Each model is trained using the same training dataset and evaluated using the same testing dataset to ensure a fair comparison.
- The same input features and target variable are used across all models, ensuring consistency throughout the experiment.
- Hyperparameters for Decision Tree and Random Forest models are tuned by adjusting parameters such as maximum depth, minimum samples split, and number of estimators to improve prediction accuracy.
- Cross validation is performed to evaluate model stability and reduce the effect of variations caused by a single train test split.
- Model performance is measured using Mean Absolute Error (MAE), Mean Squared Error (MSE), Root Mean Squared Error (RMSE), and R^2 Score.
- The experiment design ensures that preprocessing steps are applied consistently to all applicable models before training.
- Data leakage is prevented by restricting all model training, preprocessing, and parameter tuning to the training dataset only.
- The final model is selected based on its overall performance, generalization capability, and comparison with the baseline and other regression models.

7. Evaluation Results

7.1. Evaluation Findings

- Five regression models were evaluated, namely Simple Linear Regression, Multiple Linear Regression, Polynomial Regression, Decision Tree Regression, and Random Forest Regression.
- Model performance was assessed using Train RMSE and Test RMSE to determine prediction accuracy and the ability of the models to generalize to unseen data.
- Random Forest Regression achieved the lowest Test RMSE of 49,075.70, making it the most accurate predictive model among all the models developed.
- Multiple Linear Regression recorded the smallest generalization gap of 744.47, indicating consistent performance between the training and testing datasets.

- Decision Tree Regression produced moderate prediction accuracy with a Test RMSE of 75,099.37 and a generalization gap of 2,772.68.
- Polynomial Regression and Simple Linear Regression produced comparatively higher prediction errors, indicating lower predictive performance than the ensemble model.
- The comparison between Train RMSE and Test RMSE served as the diagnostic analysis for identifying overfitting and underfitting. Models with a smaller gap demonstrated better generalization capability.
- Segment wise error analysis indicated that Random Forest Regression consistently produced lower prediction errors across different housing price ranges compared to the remaining models.
- Based on the evaluation results, Random Forest Regression was selected as the final model because it achieved the highest prediction accuracy despite having a larger train test performance gap.
- The evaluation confirms that ensemble learning methods provide superior prediction performance for the California Housing dataset when compared with individual regression models.

7.2. Train and Test Performance Comparison

Model	Train RMSE	Test RMSE	Generalization Gap
Simple Linear Regression	83,418.33	84,976.53	1,558.20
Multiple Linear Regression	69,411.66	70,156.12	744.47
Polynomial Regression	83,047.76	84,652.79	1,605.02
Decision Tree Regression	72,326.69	75,099.37	2,772.68
Random Forest Regression	18,240.56	49,075.70	30,835.14

7.3. Model Performance Summary

Model	Overall Accuracy	Generalization	Final Assessment
Simple Linear Regression	Moderate	Good	Acceptable baseline model
Multiple Linear Regression	Good	Excellent	Stable predictive model
Polynomial Regression	Moderate	Good	Limited improvement over linear regression
Decision Tree Regression	Good	Moderate	Suitable for non linear relationships
Random Forest Regression	Excellent	Good	Best performing model

7.4. Diagnostic Analysis

Diagnostic	Observation	Interpretation
Train vs Test RMSE	Compared for every regression model	Used to evaluate model generalization

Generalization Gap	Smallest for Multiple Linear Regression	Indicates stable performance on unseen data
Prediction Accuracy	Highest for Random Forest Regression	Lowest prediction error among all models
Model Selection	Random Forest Regression	Selected as the final predictive model

8. Interpretations

- Median Income emerged as the most influential feature across the developed regression models, indicating a strong association with median house values.
- Geographical attributes such as latitude, longitude, and ocean proximity also contributed to prediction performance, suggesting that location related information is valuable for estimating house prices.
- Multiple Linear Regression provides interpretable coefficients, allowing the user to understand the direction and relative magnitude of each feature's influence on the predicted house value.
- Tree based models, particularly Random Forest Regression, determine feature importance by evaluating how much each feature contributes to reducing prediction error during model training.
- Random Forest Regression achieved the highest prediction accuracy, demonstrating its ability to capture complex and non linear relationships within the dataset.
- Multiple Linear Regression showed the smallest generalization gap, indicating stable performance and good consistency between the training and testing datasets.
- Decision Tree Regression captured non linear patterns effectively but showed lower prediction accuracy than Random Forest Regression.
- Simple Linear Regression produced the lowest predictive performance because it relied on a single explanatory variable and could not represent the complexity of the housing dataset.
- The experiment highlights the trade off between model interpretability and prediction accuracy. Linear regression models are easier to interpret, whereas ensemble models such as Random Forest provide higher predictive performance at the cost of reduced interpretability.
- The observed feature influences and model behavior indicate statistical associations with house prices. These findings should not be interpreted as evidence that any individual feature directly causes changes in house values.

9. Limitations And Risks

- The dataset is based on the 1990 California Census, which means it does not represent current housing market conditions or recent economic changes.
- The dataset covers only California, limiting the model's ability to generalize to housing markets in other states or countries.
- The target variable, median_house_value, is capped at the maximum value recorded in the dataset, which may reduce prediction accuracy for high value properties.
- The dataset does not include important real estate factors such as property condition, crime rate, school quality, interest rates, or recent market demand, which may influence house prices.
- The Random Forest Regression model achieved the lowest prediction error but showed a relatively large generalization gap, indicating a possibility of overfitting compared to the other models.
- Changes in housing market trends over time may lead to data drift, reducing the model's prediction accuracy if applied to newer housing data without retraining.
- Bias may exist because the dataset represents only one geographical region and one time period, which may not capture the diversity of housing markets.
- Predictions generated by the model contain inherent uncertainty and should be treated as estimates rather than exact property values.
- The model should not be used as the sole basis for financial investments, property valuation, mortgage approval, insurance assessment, or legal decisions without additional expert evaluation.
- The experiment is intended for educational and research purposes to demonstrate regression techniques and comparative model evaluation rather than for direct commercial deployment.

10. Future Improvements

- Collect more recent housing data to ensure that the model reflects current market conditions and improves prediction accuracy.
- Expand the dataset to include multiple states or countries so that the model can generalize better across different housing markets.
- Incorporate additional features such as property condition, crime rate, school ratings, interest rates, employment statistics, and nearby amenities to improve predictive performance.
- Perform spatial validation by training and testing the model on different geographical regions to evaluate its performance across locations.
- Apply temporal data splitting, where older data is used for training and newer data is used for testing, to better simulate real world prediction scenarios.
- Estimate prediction uncertainty by generating confidence intervals or prediction intervals, allowing users to understand the reliability of each prediction.
- Explore advanced machine learning algorithms such as Gradient Boosting, XGBoost, LightGBM, and CatBoost to determine whether they outperform the current Random Forest model.

- Deploy the final model as a web application or REST API to enable real time house price prediction for end users.
- Implement continuous monitoring after deployment to detect data drift, performance degradation, and changes in housing market patterns, followed by periodic model retraining.
- Establish an automated machine learning pipeline that supports data validation, model retraining, performance evaluation, and version control, ensuring the model remains accurate and reliable over time.

11. Conclusion

The present experiment successfully demonstrates the complete process of developing and evaluating machine learning regression models for house price prediction using the California Housing dataset. The dataset was inspected, analyzed, preprocessed, and used to train multiple regression models under a consistent experimental framework.

Performance evaluation showed that all developed models outperformed the baseline approach, with Random Forest Regression achieving the best prediction accuracy. Although the model exhibited a larger generalization gap than some other models, it produced the lowest testing error and the most reliable predictions overall.

Therefore, the stated objective of accurately predicting median house prices was achieved. The experiment also highlighted the importance of data exploration, model comparison, validation, and performance evaluation in selecting an appropriate predictive model.

Overall, the developed model satisfies the success criteria established for the experiment and provides a dependable solution for educational purposes while demonstrating the practical application of regression techniques in predictive analytics.